Nineteen Predictions, One Discriminator, and a Reproducibility Standard

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Abstract

This note collects the falsifiable predictions of the rope hypothesis in one place, for outside review. The hypothesis is a non-standard, self-published framework; nothing here asks the reader to accept its ontology. Each prediction is a stated mathematical relation with its inputs classified as derived or measured, its test named, and its falsifier spelled out. The inventory now stands at fifteen: four testable numbers (Part I), two structural consequences (Part II), three predictions from strand extensibility and molecular geometry (Part III), and the six-member measurement-sector family added at v2.2.5 from the strand-level campaign (Part IV), plus one declared retrodiction. This document supersedes all earlier prediction notes; it is the single inventory, maintained under one standard.

1. How to Read This Note

A prediction is only as good as its honesty about what stands behind it. For each prediction below we separate three things a reviewer needs:

The number or shape — the specific quantity or structure the framework commits to, with no remaining freedom to adjust.

The inputs — stated explicitly as DERIVED (forced by the rope structure) or MEASURED (taken from experiment as a fixed input). A prediction that leans on a measured input is still falsifiable, but it is not parameter-free, and we say so.

The test and the falsifier — the experiment that will decide, and the result that would kill the prediction.

None of these predictions depends on accepting the rope ontology. Each follows from a stated mathematical relation; a reader may accept, test, or reject any one independently of the others and of the framework that motivated it. One caveat governs several entries in Part IV: the framework has no derived absolute scale, so those entries commit to SHAPES and RELATIONS while leaving locations in frequency or length open; each such entry is marked scale-open.

2. The Predictions at a Glance

#	Prediction	Committed number / shape	Test (arena)	Status / horizon
1	MOND acceleration	1.04×10^{-10} m/s^2	H_0 convergence; z-drift of g†	DESI/Euclid, ongoing

#	Prediction	Committed number / shape	Test (arena)	Status / horizon
	scale $g^\dagger = cH_0/2\pi$			
2	Neutrino mass sum $\Sigma m\nu$	58.5 meV	CMB + BAO + galaxy surveys	CMB-S4/ Euclid, ~2027+
3	Cosmic birefringence EB/EE	0.0119, flat in ℓ	CMB EB cross-power	LiteBIRD, ~2030s
4	MOND interpolation shape	(retired)	Galaxy rotation curves	RETIRED at GRV-031
5	Gauge-boson Koide structure	structural	(no direct test yet)	—
6	α–G covariation, $d \ln \alpha = -2\, d \ln G$	ratio −2 (provisional)	quasar $\dot{\alpha}$ + LLR/pulsar $\dot{G}$ bounds	available now (bounds)
7	Vacuum Kerr nonlinearity	3:1 anisotropy, negative Δn	vacuum birefringence (PVLAS-class)	confronted; window open
8	Dark longitudinal tension channel	superluminal, cubic-only coupling	stress-channel searches	structural
9	Heavy-hydride 90° asymptote	H_2Po 90.0–90.7°; BiH_3 90.0–91.5°	molecular spectroscopy	available now
10	Gapped weave noise floor	gap + band $\times\sqrt{(1+4kt)}$; no soft modes	decoherence spectroscopy	scale-open
11	Detection = nucleation kinetics	first-passage package	threshold-detector statistics	available now
12	Spin-½ requires extension	substructure at some scale	compositeness searches	one-sided
13	CHSH cap, failed organ	$2\sqrt{2} = 2.828$	loophole-free Bell tests	standing

#	Prediction	Committed number / shape	Test (arena)	Status / horizon
	named			
14	Born exponent from channel energy	p = 2	threshold response calibration	available now
15	Analog universality (cliff + coverage)	slope ≈ −8.5/width; crossover at contact size	pendulum/cold-atom/ domain-wall analogs; filament brushes	available now
16	Isolated holes do not evaporate	(silence)	GRV-039	REGISTERED DIVERGENCE; PBH-burst falsifier armed
17	The whisper:	kappa-law derive twice; coefficie contested in-hou (0.23 vs 2e-6..3 1.3 kappa/2pi	d GRV-040-044, nt GRV-087-088 se e-5);	Modeled +

Update at v3.7.0 and after: the measurement-sector family (Part IV) resolved into a TWO-REGIME picture through the strand-kinetics campaign (FND-STRAND-009..021). Prediction 11 now stands in final form (steady-state Poisson with the band-gap prefactor; a switch-on transient after a quench), and three new entries join the list: Prediction 23 (the switch-on fingerprint: dark-count drift at constant temperature), Prediction 24 (the parameter-free detector size law, the corpus's newest DERIVED result), and Prediction 25 (Predictions 10 and 11 pinned to one spectral scale). Predictions 23 and 24 are the family's most experimentally distinctive members: existing single-photon hardware, statistics campaigns only, with the constant-temperature drift and the zero-parameter power law as signatures no rival drift model supplies.

Entries 5 and 8 are structural consequences stated for completeness rather than near-term numbers; entry 6 is provisional on one derivation, as its section states; scale-open entries commit to shape while the absolute scale remains underived; entry 12 is one-sided in practice. Entries 10–15 were added at v2.2.5 from the strand-level measurement campaign.

Part I — Four Testable Numbers

READER'S NOTE — the discrimination audit (added 1 August 2026)

This paper lists nineteen predictions. A systematic audit on 1 August 2026 asked a narrower question of each: does its OBSERVABLE OUTCOME differ from what standard physics predicts, is it checkable, and is it live? Exactly ONE qualifies on all three. Readers should not take the count of nineteen as a count of ways this framework could be distinguished from standard physics.

THE ONE: Prediction 6, d ln alpha = -2 d ln G. Confronted against published optical-clock and lunar-laser-ranging data (four independent alpha determinations, three independent G determinations) it holds at 1.74 sigma -- a null-versus-null consistency, so survival rather than confirmation. Its substance is the inverse commitment Gdot/G = -5.0(5.5)e-19 per year; any firm detection above 1.6e-18 per year refutes it. The named weak point is PSR J1713+0747, already 2.06 sigma against the relation on its own; a 3 sigma confirmation of its 2025 central value kills the claim and needs only a factor 1.45 in sigma, reachable around 2027-2030.

WHY THE OTHERS DO NOT DISCRIMINATE, in the audit's own terms:

Prediction 3 (cosmic birefringence) is CONFIRMED -- Planck legacy data favours the constant-angle model by Bayesian evidence -- but the standard ultra-light axion predicts a constant angle too. The frequency scaling that would separate them was derived and closes the door: material optical activity gives beta ~ nu^2, excluded at 4.9 sigma and overshooting the observed angle by 7e18 at the framework's own strand scale. Only the axion's nu^0 survives, and it must be imported as a postulate.

Prediction 2 (neutrino sum 58.5 meV) is sharply falsifiable -- inverted ordering or a sum away from 58-59 meV kills it -- but sits where any minimal-normal-ordering model lands, so a confirming measurement would not select this framework.

Predictions 16 and 17 (black holes) are distinctive but not reachable. Hawking radiation has never been observed from any astrophysical hole, and the only channel where evaporation is observable needs a ~1e15 g primordial hole not known to exist. The whisper at 0.23 kappa is genuinely sourced in the gravitational channel -- a reconnection is purely deviatoric with 93 percent of its excited power in the Einstein-Hilbert channel -- but it is CHANNEL-SATURATED rather than supply-limited: the horizon at 0.23 kappa is a deep-sub-wavelength emitter, so the luminosity saturates at 5.8e-4 times the Hawking power, mass-independently -- about 5e-34 W for a stellar hole, giving a strain of order 1e-58 at 10 kpc. That is Hawking-faint by construction and NO DETECTION CLAIM IS MADE. The framework's OWN work closes it twice over: the spectrum is broadband quasi-thermal rather than a line, which forbids coherent integration, and the channel ceiling sits thirty orders below even that. See GRV-047 and GRV-053.

Prediction 19 (polarimetry) is an ACCEPTANCE TEST, not a prediction: the branch that supports the framework is a QED-like result, which discriminates nothing since QED already predicts QED, and the branch that would discriminate requires a non-spinor rope electron the corpus has not built. It remains a valuable standing commitment -- if that electron is ever built non-spinor, the framework dies.

Prediction 18 (medium rest frame) commits no specific observable, and its derivation inherits a relation retired with the mesoscopic-hbar sector.

Prediction 1 (the MOND scale g+ = cH0/2pi at zero parameters) is a real achievement and does NOT discriminate, because MOND fits the same observable.

A CORRECTION TO PREDICTION 19's SCOPE. The polarimetry axis was registered as the arbiter for the vacuum stiffness Sigma. It is not. Discriminating the two registered Sigma candidates needs a birefringence sensitivity roughly 5e8 beyond VMB@CERN's design goal. The axis is a THRESHOLD test on Sigma's branch (already passed, since PVLAS excludes the small-Sigma regime) and a genuine near-term test of the ELECTRON's internal class. Sigma's value is settled by calculation, not by that experiment.

WHAT THE AUDIT DOES NOT SAY. None of the above means these predictions are wrong, and several remain genuinely falsifiable even where they cannot discriminate: the neutrino sum dies on inverted ordering, the polarimetry commitment dies on a non-spinor electron, the black-hole claims die if an evaporating primordial hole is found. What the audit says is narrower and should be stated plainly: this framework currently offers ONE way to be distinguished from standard physics, and several ways to be killed. The corpus also carries twenty-three registered Failed-and-kept claims, which is the more informative number about whether it can be falsified. Full reasoning: ELEC-062, ELEC-063, ELEC-064, GRV-049, GRV-052, PRED-002-CONF, PRED-002-FREQ, PRED-003-CONF, PRED-003-META, PRED-003-J1713, QGATE-018.

Prediction 1 — The MOND Acceleration Scale

The relation. The MOND characteristic acceleration $g^\dagger \approx 1.2\times10^{-10}$ m/s^2 is an empirical parameter in Modified Newtonian Dynamics and in the Radial Acceleration Relation observed in disk galaxies. The rope framework proposes it is not free but fixed by the Hubble rate:

$$\mathbf{g^\dagger = c\,H_0 / 2\pi = c^2 / (2\pi\, R\_H)}$$

connecting the galactic acceleration scale directly to cosmology using only c, H_0, and a geometric factor 2π. No galaxy-specific fitting enters.

Inputs. MEASURED: H_0 (the Hubble constant); the prediction inherits the Hubble tension — an irreducible ~9% spread (H_0 = 67–73 km/s/Mpc) until H_0 is pinned. DERIVED (partially): the form $cH_0/2\pi$ from the horizon coupling. HONEST CAVEAT: the 2π is motivated by horizon-sphere geometry but is not uniquely forced; the alternative $cH_0/6$ performs comparably against SPARC (RMS 0.203 vs 0.201). We do not claim the coefficient is settled.

Test and falsifier. With current local H_0, $g^\dagger = cH_0/2\pi$ sits about 6% below the SPARC value — right ballpark, not confirmed. The sharper, framework-specific test is drift: $g^\dagger$ and H_0 should track together across cosmic time. A measurement of $g^\dagger$ from high-redshift galaxy samples that does NOT scale with H(z) would falsify the relation; so would H_0 converging far from the value that makes the relation hold, with no z-drift compensating.

Prediction 2 — The Sum of Neutrino Masses

The relation. The Koide formula relates the three charged-lepton mass amplitudes to a phase on a circle; the measured lepton masses fix that phase at $\varphi = 132.733°$. Applying the same Koide structure to neutrinos with a fixed offset $\pi/12 = 15°$ (Brannen 2006) gives $\varphi\nu = 147.733°$. With the measured solar splitting as the one scale input, all three neutrino masses are then determined:

$\Sigma m\nu$ = 58.5 meV

Inputs. MEASURED: the solar mass splitting Δm^2_{21} (the one scale), and the ratio $\Delta m^2_{31}/\Delta m^2_{21}$ = 32.6 used to fix the offset. INPUT, NOT DERIVED: the $\pi/12$ offset itself. HONEST CAVEAT: $\pi/12$ is an empirical choice, not derived from the rope structure. A skeptic who treats the offset as fitted will regard $\Sigma m\nu$ as a consistency check rather than a pure prediction. We state it as the former.

Test and falsifier. $\Sigma m\nu$ = 58.5 meV is the one number here not yet measured, with nothing left to adjust. It sits just above the normal-ordering minimum (~59 meV), making it sharp and risky: distinguishable both from the quasi-degenerate spectrum (Σ ~ 100–200 meV) and, with care, from the bare splitting minimum. Falsified by a confirmed $\Sigma m\nu$ below ~50 meV or above ~70 meV, or by confirmation of inverted ordering. Cosmological bounds are already pressing toward this range.

Prediction 3 — Cosmic Birefringence (EB/EE)

The relation. The rope is a chiral object: its two strands wind with a definite handedness. That chirality acts on propagating light as a Chern-Simons-type coupling, rotating the plane of linear polarization as it travels — cosmic birefringence. Integrated to the last-scattering surface, the rotation angle is $\beta = \chi cD/2$ with $\chi = \sin(2\theta\_W)$ n_rope r_H^2/c. Eskilt & Komatsu (2022) report $\beta = 0.342° \pm 0.094°$ from Planck; matching that fixes the rope chirality product to $\sin(2\theta\_W)$ n_rope $r\_H^2 = 9.18\times10^{-29}$ m^{-1}. The framework then predicts the CMB EB cross-correlation:

EB/EE = $\sin(4\beta)/2$ = 0.0119, flat in multipole ℓ

Inputs. DERIVED: the mechanism (a parity-odd CS coupling from the rope helix) and the functional form EB/EE = $\sin(4\beta)/2$, flat in ℓ — the flatness is a genuine rope-specific signature; many alternative mechanisms predict ℓ-dependence. MEASURED: the rotation angle β (from Planck), which fixes the chirality product; the individual geometric factors are not separately determined.

Test and falsifier. LiteBIRD (launch ~2030s) is designed to measure CMB EB to the precision this prediction requires. Falsified if EB/EE is inconsistent with $\sin(4\beta)/2$ at the measured β, or if the cross-correlation shows significant multipole dependence rather than the predicted flatness. This is the cleanest near-term test of a rope-specific mechanism.

Prediction 4 — The MOND Interpolation Shape [RETIRED at GRV-031]

RETIRED at GRV-031 (July 2026), kept visible per house rules. The original claim -- a tanh-type interpolation shape differing ~8 percent from standard forms -- was never pinned to an exact nu(y) (the audit finding of GRV-030), and the corpus-natural derivation candidate (Langevin alignment-saturation against the horizon floor, parameter-free) was stated, bar-locked, and tested on 2788 SPARC points: it loses to the simple interpolation by 0.015 dex at the predicted g_dagger. Both the tanh family and the derived candidate are disfavored; the shape prediction is retired. The framework's registered MOND content is Prediction 1 -- the acceleration scale, confirmed at zero free parameters on the same dataset -- carried by the horizon channel alone (no galaxy-scale scalar exists; GRV-028/029). Any future shape candidate must beat 0.1421 + 0.003 dex at the predicted g_dagger to enter this paper.

Part II — Structural Consequences

Prediction 5 (Structural) — Gauge-Boson Koide Structure

At the rope's Chern-Simons level k = 3, the quantum dimensions of the spin-½ and spin-1 representations coincide ($d_{1/2} = d_1 = \Phi$, the golden ratio, because $\sin(2\pi/5) = \sin(3\pi/5)$). This forces the W and Z bosons onto the SAME phase structure $\varphi = (3+\Phi)\theta_W$ that governs the charged leptons — a lepton/gauge-boson link the Standard Model does not contain. The underlying identity is exact mathematics. Stated as structural only: a clean consequence without a direct experimental test at present, and not counted among the four numbers of Part I.

Prediction 6 (Structural / Provisional) — Correlated Variation of α and G

The relation. The electromagnetic and gravitational couplings are expressed in the SAME rope-medium primitives: $\alpha \sim 2T^2/(\kappa a)$ from the EM energy coefficient, and G governed by the same tension T. Because both are functions of one shared set of primitives, α and G are NOT independent — a sharp departure from standard physics. If a shared primitive drifts, they co-vary in a fixed, scale-free ratio; for tension-driven drift:

$d \ln \alpha = -2\, d \ln G$ (equivalently $\dot{\alpha}/\alpha = -2\, \dot{G}/G$)

Inputs. DERIVED: $\alpha \sim 2T^2/(\kappa a)$, whose chain is derived to rope-medium primitives with the locking energy $J = T^2/\kappa$ exact in the harmonic regime. ASSUMED: $G \sim 1/(Ta)$, the natural tension-rigidity scaling, NOT yet derived to the same rigor — the specific exponent −2 depends on it.

Test and falsifier. Testable now against combined bounds: quasar-absorption limits on $\dot{\alpha}$ and lunar-laser-ranging / pulsar-timing limits on $\dot{G}$. A measured drift ratio inconsistent with −2 (for tension-channel drift) falsifies the shared-tension structure. Two qualifications: (i) the −2 is PROVISIONAL on deriving $G(T,\kappa,a)$ rigorously — the framework-forced content is a FIXED ratio, not the specific value; (ii) current data are consistent with no variation of either constant, so this is a bound and a standing prediction, not a detection.

Part III — Extensibility and Molecular Geometry

Prediction 7 — Vacuum Kerr Nonlinearity, Quantified and Confronted

The relation. The strand stretch modulus k, required for the repulsive core (EM-RECON-009), makes light self-interact at quartic order with onset strain $g^* = \sqrt{4T_0/(k-T_0)}$ (EM-RECON-010). The polarization structure is computed exactly: anisotropy ratio 3:1 with NEGATIVE index shifts (background fields stiffen the network; light speeds up), versus QED's 7:4 with positive shifts — a double qualitative discriminator (EM-RECON-016).

Confrontation to date. Matching the magnitude against the final PVLAS limit EXCLUDES identifying the rope quartic with the observed light-by-light nonlinearity (overshoot ~570×) and sets $\Sigma \gtrsim 8.6\times10^{27}(k/T_0-1)$ J/m³ — the framework has already been corrected once by this confrontation, and says so. Decisive window: vacuum-birefringence experiments improving toward QED sensitivity will either detect a negative Δn with 3:1 anisotropy (rope signature) or push Σ higher and confirm QED. Either outcome is informative.

Prediction 8 (Structural) — A Dark Longitudinal Tension Channel

Stability requires $k > T_0$, so longitudinal strand waves propagate at $c_{L,eff} = \sqrt{(k+\lambda\gamma^2\tau_0^2)/\mu}$ > c, with $\gamma = 1/\sin^2\theta$ from helix geometry (EM-RECON-012). The channel decouples from matter and light at linear order exactly (cubic-only coupling) and creates no causal paradox in the emergent-Lorentz framework (EM-RECON-011). SHARPENED AT v3.12 (FND-027/028): with the k/T_0 dispute adjudicated to the single registered value $k/T_0 = 2$, the channel speed acquires a committed one-sided floor, $c_L \geq \sqrt{2}\,c = 1.414\,c$, with equality when the twist-stretch stiffening $\lambda\gamma^2\tau_0^2$ is negligible against T_0 — and the cubic coupling coefficient is now committed at $(k-T_0)/2 = T_0/2$, finite and O(1) in tension units (superseding an earlier remark that it grows without bound, which described the inextensible limit the adjudication closed). The floor and the nuclear/chemical core coefficient $c_4 = (k-T_0)/8 = T_0/8$ are set by the SAME constant: one number, two sectors, zero freedom between them. Falsifiers, sharpened: any determination of the longitudinal channel speed below $\sqrt{2}$ c; any observed LINEAR coupling to the channel; or a spacing-sector re-determination forcing k/T_0 far from 2, which breaks the cross-sector lock.

Prediction 9 — The Heavy-Hydride 90-Degree Asymptote

The relation. The molecular-geometry theorems make 90 degrees the raw fixed point for single bonds on orthogonal dipolar lobes, approached from above as the positive opening corrections collapse down each column. Confirmed on eight hydrides (H_2O 104.48° → H_2Te 90.25°; NH_3 106.67° → SbH_3 91.70°): monotone, one-sided (a single 89-degree hydride would have falsified the structure), with the $XH_3 > XH_2$ crowding law holding at every period.

Standing predictions, registered in advance: H_2Po between 90.0° and 90.7°; BiH_3 between 90.0° and 91.5° — one-sided windows. Tier label: consistency-tier against quantum chemistry (which predicts these angles too); the distinctive content is the asymptote's derivation from two integration-verified theorems rather than a hybridization narrative.

Part IV — The Measurement-Sector Family (added at v2.2.5)

With the v2.2.5 cycle, the measurement chain — amplitude, detection event, reservoir, and reservoir spectrum — was constructed on literal strands (FND-STRAND-005 through 008), and a family of falsifiable commitments follows. Scale-open entries commit to shapes and relations; the locations remain underived.

Prediction 10 — The Gapped Environmental Noise Floor (scale-open)

The relation. The weave reservoir's normal-mode spectrum is gapped — every mode carries the strand mass, band [ω_{gap}, $\omega_{gap}\sqrt{(1+4kt)}$], NO soft modes; environmental noise sourced by the fundamental medium arrives only at and above the gap. DERIVED: the gap's existence and the band shape, from the engine's own Hessian (FND-STRAND-008; matched to dispersion at 7×10^{-16}). OPEN: the gap's location in hertz. Falsifier: an established, irreducible, gapless environmental noise spectrum extending to zero frequency at every accessible scale, attributable to the fundamental medium rather than to ordinary matter environments. KINETIC COROLLARY (v3.7.0, FND-STRAND-010/013): the same gap is the CLOCK of detector kinetics — the Kramers attempt rate in Prediction 11's steady-state dark-count law sits at the gap frequency to O(1), confirmed on two estimators with opposite failure modes. Two instrument-facing predictions, one spectral location: a rigidity a flexible framework could not manufacture.

Prediction 11 — Detection Is Nucleation Kinetics

The relation, in its FINAL TWO-REGIME FORM (the strand-kinetics campaign, FND-STRAND-009..021, v3.7.0 and after). The detector click is a thermally activated topological event (kink-pair nucleation, FND-STRAND-006/007). STEADY STATE: at every detector size, clicks are Poisson — exponential waiting times at the stationary Kramers rate, Arrhenius in temperature, with the attempt-rate prefactor at the weave band gap (the strand mass scale; FND-STRAND-010, estimator-robust per FND-STRAND-013) and latency decreasing with channel energy. AFTER A QUENCH (power-up, reset, sudden isolation): the click rate starts elevated and relaxes to the steady value over one bath-correlation epoch — product-state initial slip (FND-STRAND-021), measurable as a dark-count rate drifting DOWN while the device temperature holds CONSTANT, a signature no thermal-drift model reproduces. The quench transient's size dependence is an exact, parameter-free law (Prediction 24). SUPERSEDED WITHIN THIS PREDICTION: the earlier open-ended non-exponential phrasing, and an interim mechanism-conditional drift claim retired at FND-STRAND-016 — the registry records both. DERIVED: the event's existence, localization, irreversibility, channel-energy drive, and (at Derived status) the size law. HONEST CAVEAT: parts of the steady-state package resemble known avalanche-detector phenomenology; the commitment is to the WHOLE package including the quench regime, which does not. Falsifiers: steady-state click statistics that are non-activated or non-Poisson where the model's detector physics applies; OR a small isolated detector that shows NO quench transient under conditions where the model requires one; OR a quench transient whose decay tracks device temperature rather than persisting at constant temperature.

Prediction 12 — Spin-½ Requires Extension (one-sided)

The relation. The half-angle response is the SU(2) lift, which a point orientation cannot carry and a framed strand cannot avoid (FND-STRAND-005; measured at 10^{-9} for arbitrary axes and profiles). Commitment: every spin-½ particle has internal structure at SOME finite scale. Scale-open, hence one-sided in practice: compositeness bounds may rise indefinitely without contradiction, but a CONFIRMED structureless point with spin-½ at all scales would falsify the mechanism outright.

Prediction 13 — Tsirelson Forever, with the Failed Organ Named

The relation. CHSH-type correlations never exceed $2\sqrt{2}$ — shared with quantum mechanics; distinctive here is the WHY and therefore the autopsy: supra-Tsirelson statistics require a setting-dependence channel of polynomial degree greater than 24 where this framework's physics has degree exactly 1 (QB-018/019/022). Falsifier: any confirmed loophole-free violation above $2\sqrt{2}$ — which kills the measurement sector at a named claim, not merely embarrasses it.

Prediction 14 — Born's Square from Channel Energy

The relation. The quadratic ($p = 2$) response near detection threshold is the statement that nucleation is driven by channel ENERGY, quadratic in amplitude (QB-011 protocol; FND-STRAND-005 B4, $p = 1.99$). Falsifier: a confirmed, reproducible deviation from quadratic threshold response — breaking the chain at a specific link (the energy-drive identification), leaving the rest standing for inspection.

Prediction 15 — Analog Universality (testable now)

The relation. Any engineered system realizing the twist-chain physics — coupled pendulum arrays, magnetic domain-wall lattices, cold-atom sine-Gordon simulators — must exhibit the friction cliff (seven-order exponential mobility suppression, slope ≈ −8.5 per width unit, arrest/coast dichotomy) with the LAW transferring across implementations and only prefactors carrying the coupling convention (FND-KIN-002; FND-STRAND-002, slope agreement 0.16%). Engineered filament brushes should likewise show the transparent-to-solid coverage crossover at gap ≈ contact size with the measured inverse-fourth-power approach (FND-STRAND-004). No knowledge of the absolute strand scale is required, because the experimenter builds the system.

A retrodiction, declared as such: the Călugăreanu ledger and the supercoiling instability (FND-STRAND-003) are confirmed daily by DNA mechanics — consistency evidence for the strand engine, counted here as retrodiction, not prediction.

Honest Status of the Framework

For any reviewer evaluating these predictions, the following should be stated plainly:

The rope hypothesis is non-standard, self-published, and not peer-reviewed. These predictions are presented as empirical consequences of specific stated relations, not as derivations from established physics.

Two of the four testable numbers (the neutrino $\pi/12$ offset; the MOND 2π coefficient) rest on inputs that are empirical choices, not derived. They are falsifiable but not parameter-free, and we have marked exactly where.

The framework's deepest open problem — deriving absolute particle masses (and hence atomic and molecular masses, and the absolute strand scale that would close it) — now has a campaign and a receipt trail (see: The Mass of Matter Gets a Receipt). The mass model m = Sigma L + lambda [2 kc x contacts + bend(kappa)] has its structural terms fully derived: the contact term is an exact parity theorem and the bend law is closed with zero free parameters (dE x L = -0.509658 + 2 pi (5/2 - 7 sqrt2/4) ln L). The sector has since closed most of that gap from inside. The contact stiffness kc was DISSOLVED (the contact is a constraint, not a spring: an exact parity theorem, with the binding sign derived), and the scale lambda was taken through a full internal arc: collapsed by two identities (beaded string; waves at c) to one constant; its quantum reading KILLED by the framework's own Lorentz bound (an internal, arithmetic audit that also disfavored the doublet-as-n/p hypothesis in that reading); its surviving classical-amplitude reading made exactly extensive by a trace identity; its last parameter confiscated by collision saturation and closed by the density-onset principle at the tube's internal bundle (x equals the square root of the coverage threshold); and finally COMPUTED by percolation simulation: Lambda = 0.013 to 0.017 with zero free parameters, landing at the low edge of the phenomenological window. The conditioning sector is parameter-free -- m = T D (L - Lambda ell) -- and the single remaining scale input is T D itself, which now carries the absolute-mass question alone. The 1836.15 fork stands; every hypothesis-shaped statement remains marked; and the walls of the prototype solver, terminal-graded and congestion-gauged, gate the precision of every conditional until a production-grade solver arrives.

The strongest internal result — lepton mass RATIOS from $\varphi = (3+\Phi)\theta\_W$ to ~0.5% — is a CONJECTURE, not a theorem, and is not listed among the numbers above.

Within the measurement sector, two grants remain and are named wherever they matter: the pair is one thing (Bell-protected), and the weave is warm (a temperature, not a law).

Acknowledgment of Origin

The core idea underlying this work — that physical phenomena arise from a two-strand rope-like structure connecting matter, with light as a transverse disturbance along the rope, and the principle that physical theories must describe objects rather than non-object concepts — originates with Bill Gaede (the "Rope Hypothesis" / "Thread Theory"). The predictions and the computational, field-theoretic, and topological development presented here are an independent, modified mathematization that departs from his formulation in substantive ways. Crediting this origin is not a claim that Gaede endorses this work, nor an endorsement of all of his positions; it is the honest acknowledgment of where the foundational idea came from.

References

Eskilt, J. R. & Komatsu, E. 2022, Phys. Rev. D 106, 063503 (cosmic birefringence).

Brannen, C. 2006, unpublished (Koide neutrino phase offset π/12).

Koide, Y. 1983, Phys. Lett. B 120, 161 (charged-lepton mass relation).

Lelli, F., McGaugh, S. & Schombert, J. 2016, AJ 152, 157 (SPARC database).

Software: rope-framework v2.2.5 (open source; DOI 10.5281/zenodo.21430784) — frozen benchmark IDs; every number in this note reproducible from the registry (claims.yaml) and its 155 passing benchmarks.

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Part V — The Black-Hole Discriminators (added at v2.2.8)

Prediction 16 — Isolated Black Holes Do Not Evaporate

The corpus's horizon is a percolation-severed transverse channel (GRV-035): steady Hawking flux requires continuous vacuum conversion at an infinite-redshift surface or a supply flow, and the derived structure provides neither -- measured directly in the formation experiment (GRV-039: the collapse radiates a directed burst, after which the steady current settles to about one percent of the Stefan-Boltzmann prediction). PREDICTION: black holes emit when made and when fed; an isolated hole is eternally quiet, and primordial black holes do not undergo terminal evaporation. FALSIFIER, in capitals in the registry: A DETECTED PRIMORDIAL-BLACK-HOLE EVAPORATION BURST WOULD FALSIFY THIS BRANCH. For every observed astrophysical hole the prediction is indistinguishable from the standard picture (whose 1e-8 K is unobservable regardless); the fork is exclusively at the unfed and the primordial. Status: registered divergence (GRV-039), Modeled.

Prediction 17 — The Whisper: Accretion-Powered Emission with a Running-Temperature Tail

Fed black holes emit reconnection noise from the marginal shell: frequency scale 0.23 kappa (the strand scale cancels -- GRV-040; within a factor of two of the Hawking peak), luminosity gated by feeding as a SWITCH (GRV-047 revision: the channel-limited ceiling L = 5.8e-4 x P_Hawking, mass-independent; the proportional regime is unphysical; fed holes hum at a universal Hawking-faint level, ~5e-34 W stellar -- no detection claim is made), spectral family Planck-like with fitted temperature 1.3 times kappa-over-two-pi (GRV-043; the O(1) offset recorded as unexplained, no exact claim) -- and, the sharp part, a tail whose effective temperature RUNS: T_eff proportional to 1/omega beyond the shell edge, in closed form (GRV-044, Derived), leaving the spectrum tail-deficient relative to a matched Planck curve. Hawking's temperature is constant; the whisper's runs. The line shape is therefore a power-source discriminator even where both spectra look thermal. Scales: ~180 Hz for ten solar masses; the LISA band for Sagittarius A*. Honest limits: the efficiency question is dissolved -- f is a ratio, ~4e-67 at Eddington (GRV-047); the kernel is 1+1; amplitude separability from ordinary accretion noise is an open question, filed. Status: GRV-040-043 Modeled, GRV-044 Derived; bars were locked before every measurement in the chain. UPDATE (v3.5.0): the frequency LAW is now derived from a second, independent route — the strong-field mechanism chain (statics to ratchet thermodynamics, GRV-077..087) produces T_inf proportional to kappa by exact cancellation — but the two routes DISAGREE on the coefficient by roughly four orders of magnitude (mechanism 2e-6..3e-5 vs this entry's 0.23, GRV-088), a prediction-meets-prediction tension registered at full strength with suspects pre-listed on both chains. Readers should treat the 0.23 value and the ~180 Hz stellar figure as STANDING – the coefficient campaign has adjudicated (GRV-089..091): the tension was a CATEGORY ERROR, a temperature coefficient set against a ringing-frequency coefficient. The emission is impulsive (discrete reconnection snaps ring the throat at its resonance regardless of rate – Campbell’s theorem), so the pitch is geometric and stands at 0.23 kappa, while the mechanism’s T_inf = C hbar kappa governs the EVENT RATE. Both chains vindicated at their own questions; the proportionality to kappa, the switch-class luminosity, and the running-tail discriminator are unaffected (all three are coefficient-independent).

Part VI — The Medium-Frame and Polarimetry Discriminators (added at v2.6)

Prediction 18 — The Medium's Rest Frame Is the CMB Frame

The relation. If the vacuum is a material medium it defines a rest frame, and the CMB already defines one: we move at 370 km/s = 1.23e-3 c relative to it. The framework predicts THESE ARE THE SAME FRAME. A mismatch appears as medium anisotropy at order (v/c)^2 = 1.5e-6 in any medium-coupled observable.

Inputs. DERIVED: that a material medium defines a frame at all, and the (v/c)^2 scaling of the resulting anisotropy. MEASURED: the solar-system velocity relative to the CMB.

Test and falsifier. Any medium-coupled observable searched for a dipole or quadrupole anisotropy aligned with the CMB dipole at the 1e-6 level. Killed by a demonstrated medium-frame anisotropy misaligned with the CMB dipole. Registered claim: HBAR-010. Note: this prediction survives the retirement of the mesoscopic-hbar identification (ELEC-061), because it never used that length.

Prediction 19 — Vacuum Stiffness and the Photon Quartic on One Polarimetry Axis

The relation. Consistency of additivity, Lorentz invariance and the corpus's coherence count forces the vacuum stiffness Sigma >= 5.1e35 J/m^3. Because large Sigma makes the mesh's own nonlinearity invisible at PVLAS scales, observed vacuum birefringence must instead come from the matter sector, whose low-energy photon quartic is a fingerprint of the lightest charged particle's internal class: a spinor-equivalent electron gives the (4,7)-positive structure, and any non-spinor rope electron gives a deviation at the measurable 1e-23 scale — a 71 percent change in ratio plus a sign flip.

Inputs. DERIVED: the Sigma bound from the stated consistency conditions; the (4,7) acceptance structure. NOT YET BUILT: the rope electron model itself, so the second half is a standing acceptance test rather than a completed prediction.

Test and falsifier. A VMB@CERN-class polarimetry experiment reads both quantities on one axis. THE PAYOFF IS INVERTED relative to naive expectation: a QED-like birefringence result SUPPORTS the large-Sigma branch, while a rope-signature birefringence would kill it. Registered claims: QGATE-007, QGATE-010.

Scale-chain note (added at v2.6). The vacuum stiffness above carries a registered 28 percent tension: the framework's flux-tube mass-density radius predicts 0.342 fm against 0.407(14) fm computed on the published lattice data of Baker et al., EPJ C 85, 29 (2025). Two scale sets are therefore registered in parallel — Sigma = 5.1e35 (framework) and 3.61e35 (lattice-anchored) — and the polarimetry axis above separates them. See ELEC-052, ELEC-053.

Registered Claims in This Paper

Each statement below is traceable to the machine-readable registry (claims.yaml), the corpus's single source of truth; every benchmark reruns from source.

• CHEM-GEO-002 [Derived]: HEAVY-HYDRIDE 90-DEGREE ASYMPTOTE [benchmark: benchmarks/em/heavy_hydride_asymptote.py]

• FND-STRAND-005 [Modeled]: THE INTERACTION MODEL ON LITERAL STRANDS — the framed strand is the SU(2) lift [benchmark: benchmarks/foundations/strand_interaction_model.py]

• FND-STRAND-006 [Modeled]: THE NUCLEATION EVENT ON STRANDS — the dot as topological event [benchmark: benchmarks/foundations/strand_nucleation.py]

• FND-STRAND-007 [Modeled]: THE WEAVE AS RESERVOIR — the bath derived [benchmark: benchmarks/foundations/strand_weave_reservoir.py]

• FND-STRAND-008 [Modeled]: THE WEAVE SPECTRUM FROM THE ENGINE — the bath is gapped [benchmark: benchmarks/foundations/strand_weave_spectrum.py]

• QB-019 [Derived]: MECHANISM-UNITY SELECTS TSIRELSON [benchmark: benchmarks/bell/tsirelson_selection.py]

• QB-022 [Modeled]: MECHANISM-UNITY DECOMPOSED AND GROUNDED (SOVO) [benchmark: benchmarks/bell/mechanism_unity_grounded.py]

• GRV-039 [Modeled]: ISOLATED BLACK HOLES DO NOT EVAPORATE [benchmark: benchmarks/grv/horizon_evaporation.py]

• HBAR-010 [Modeled]: THE FORMATION-ERA MEDIUM MEETS COSMOLOGY — the medium frame is the CMB frame [benchmark: benchmarks/foundations/hbar_cosmological_test.py]

• QGATE-007 [Modeled]: THE VACUUM STIFFNESS BOUND Sigma >= 5.1e35 J/m^3 [benchmark: benchmarks/qgate/vacuum_stiffness.py]

• QGATE-010 [Modeled]: THE MATTER-SECTOR BIREFRINGENCE — the polarimetry axis becomes a spin-meter [benchmark: benchmarks/qgate/matter_birefringence.py]

• PRED-001 [Modeled]: THE NEUTRINO MASS SUM = 58.47 meV [benchmark: benchmarks/foundations/pred_neutrino_mass_sum.py]

• PRED-002 [Modeled]: COSMIC BIREFRINGENCE EB/EE = sin(4 beta)/2, FLAT IN MULTIPOLE [benchmark: benchmarks/foundations/pred_cosmic_birefringence.py]

• PRED-003 [Modeled]: THE COUPLING-DRIFT RATIO d ln alpha = -2 d ln G [benchmark: benchmarks/foundations/pred_alpha_g_drift.py]

Part VII — The Strong-Field Campaign’s Additions (added at v3.5.0)

Prediction 20 — The Hawking Law-Form from Statics (internal, now adjudicable in-house)

The strong-field campaign (GRV-074..088) derived, with bars locked before every session: the horizon's support statics (load-share and staticity as theorems), an energy-honest interior (accretion emerging unasked; collapse as permanent recorded structure), and a temperature chain — pressing x thickness = barrier (the lift-over theorem); barrier x 15.67 = bit-cost (measured to three digits); bit-cost over a counted 4-to-6 modes = temperature — whose three exact cancellations deliver T_inf = (beta K h/m*) kappa: the observed temperature proportional to surface gravity, with Hawking nowhere in the inputs. The FALSIFIABLE content is unusual and precious: the corpus now owns TWO independent internal routes to one dimensionless number (the whisper's frequency coefficient), and they disagree by four orders. At least one chain carries an unearned link; the pre-listed suspects (the bit-cost's engine-parameter provenance, prime; the thickness convention; the cell-energy identification; the original mode identification) make the adjudication a bounded computation, not a judgment call. This was a theory subpoenaing itself, and THE VERDICT IS IN (GRV-089..091, August 2026): beta promoted to a universal function with its operating point measured; the pile-up eliminated by exact cancellation; the mode identification vindicated by Campbell's theorem; and the confrontation itself convicted of a category error – the emitter is crackling-noise class (cold thermodynamics, bright snaps), the pitch geometric, the rate thermodynamic, both predictions right at their own questions. The case is CLOSED; the surviving open item is the quantum of the ring (whether each snap's energy matches hbar omega_ring), which is the hbar frontier itself. [GRV-087, GRV-088; benchmarks/gravity/flux_confrontation.py, coefficient_evaluation.py]

Prediction 21 — The Lattice Fork's Eighteen-Order Lever (internal discriminator, named)

The lattice-scale survey (GRV-076) found ZERO live pins of the spacing a: the corpus's two candidate scales (Lorentz-bound 1e-16 m; Sakharov selection 1.26e-34 m — eight Planck lengths, selected by a 96-site zero-point coefficient, GRV-075) [UPDATE 2026-08-09, GRV-101: the 96-site coefficient was EXTENSIVE -- the registered zeta carried the ring size sqrt(96), and the corrected intensive chain selects a = 0.80 Planck lengths, with the covariant chain giving (0, 0.18] l_P on the positivity half-line. The PLANCK-CLASS selection stands and is stronger; the digit eight does not.] sit on an exact degeneracy line along which every registered invariant is unchanged. The fork is a single bit — is the Sakharov identification true? — and the strong-field campaign handed it its strongest lever: the Hawking-form coefficient differs between the branches by EIGHTEEN ORDERS. Whichever way the coefficient campaign resolves Prediction 20, the surviving value will also decide the fork, collapsing the corpus's last scale ambiguity. The external discriminator remains registered alongside: the Sigma-sensitive vacuum nonlinearity (PVLAS-class), on which the forks differ by 6.3e35. [GRV-074..076, GRV-088]

Prediction 16, reaffirmed with a mechanism. The isolated-hole silence (no evaporation; primordial black holes do not explode) now rests on machinery rather than absence: the ratchet requires feeding to break crossings, the emission channel is re-formation of eaten crossings, and an unfed shell has neither. The falsifier is unchanged and armed: one observed terminal PBH burst kills the branch. [GRV-081, GRV-085]

Prediction 22 — J1713 Refutability, Certified Form-Robust

The G-constitutive audit (GRV-074) certified that the binary-pulsar refutability of the corpus's coupling-drift structure survives EVERY candidate closure of the G form, at millions of clock sigmas per candidate: the decidability of PRED-003's family this decade is G-form-robust — values were conditional, the verdict is not. [GRV-074; benchmarks/gravity/G_constitutive_bracket.py]

Prediction 23 — The Switch-On Fingerprint (dark-count drift at constant temperature)

The relation. A small, well-isolated threshold detector, freshly reset or powered up, shows dark counts whose waiting times are non-exponential and whose apparent rate declines toward the steady value over one bath-correlation epoch, WHILE every stationary aggregate — device temperature, delivered noise intensity — holds constant (FND-STRAND-014..016 measured the flats; FND-STRAND-021 identified the mechanism as product-state initial slip and removed the transient by equilibrium preparation, R_eq = 0.93 at the registered plateau rate). In steady state the same detector is textbook Poisson. STATUS: Modeled (shape and relationships; absolute timescale scale-open per FND-MATTER-003). Falsifier: a small isolated detector prepared by quench that shows constant-rate exponential dark counts from the first moment at the model-relevant scale, or a transient that tracks measured temperature drift.

Prediction 24 — The Detector Size Law (parameter-free Poissonization)

The relation. For detectors prepared identically after a quench, the dark-count survival curve at size N is the reference detector's curve raised to the size ratio: S_N(t) = S_ref(t)^(N/N_ref) — the channel count cancels and the law carries ZERO adjustable constants (FND-STRAND-020, status DERIVED; verified at three sizes inside registered confidence intervals, medians to factors 1.10-1.30). Consequence: the quench fingerprint of Prediction 23 shrinks with size by this exact power law, and large detectors read Poisson at all times — arithmetic hides the per-channel memory rather than physics removing it. Falsifier: measured size dependence of quench-prepared dark-count statistics that violates the power law beyond its stated small-size domain edge (per-channel barrier relief, FND-STRAND-012).

Prediction 25 — One Scale, Two Instruments (the gap rigidity)

The relation. Prediction 10's decoherence floor and Prediction 11's steady-state dark-count attempt rate sit at the SAME spectral location: the weave band gap, the strand mass scale (FND-STRAND-008 for the floor; FND-STRAND-010/013 for the clock, O(1) on two estimators). The framework cannot move one without moving the other. Falsifier: a measured irreducible noise floor and a measured fundamental dark-count prefactor scale that are established to sit at different spectral locations — either instrument alone constrains; together they cross-check. STATUS: the relation is committed now; the shared location in hertz inherits the scale-openness of FND-MATTER-003.

Prediction 26 — No Finite-Speed Bell Cutoff (the medium's ceiling; consistency-tier, added at v3.12)

The relation. After the k/T_0 adjudication (FND-027), the mechanical medium's fastest channel tops out at $c_L \approx \sqrt{2}\, c$ (barring the uncomputed twist addition) — four orders of magnitude below the 1.38×10^{4} c floor that Bell-timing experiments impose on any finite-speed correlation carrier, and Bancal et al. (2012) exclude every finite speed regardless. The framework therefore

COMMITS: no wave in its medium carries Bell correlations, and Bell correlations exhibit NO finite-speed cutoff at any mechanically accessible speed. The nonlocal correlations belong to the guidance-flow extension (configuration-space object), not to medium mechanics; the fence between the two is now a committed number.

Inputs. DERIVED: the ceiling $\sqrt{2}$ c (from the registered $k/T_0 = 2$ via $c_L/c = \sqrt{(k/T_0)}$); the exclusion of finite carriers (Bancal, cited). MEASURED: the Bell-timing floor (Yin 2013). Tier: consistency-tier retrodiction against existing bounds ($>10^4$ c already measured), stated because the committed CEILING is new content — no earlier version of this framework could say what its medium's maximum speed was.

Test and falsifier. Any future Bell-timing experiment observing a finite-speed cutoff falsifies standard quantum mechanics AND this framework's structure simultaneously — and a cutoff in the decade around $\sqrt{2}$ c would be uniquely catastrophic here, resurrecting the mechanical carrier at exactly the speed the elasticity permits and the adjudication forbids. The framework is exposed at a precise location where a generic preferred-frame model is not.

Prediction 27 — Intensity-Squared Dark Vacuum Attenuation (scale-open, added at v3.12)

The relation. The longitudinal channel's only coupling to light is the exact cubic vertex $(T_0/2)\, u'\psi'^2$ (coefficient committed by the adjudicated branch; sympy-verified). An intense transverse (electromagnetic) field of strain amplitude g therefore sources longitudinal strain at second order (forced $u' \sim g^2$), draining energy into the dark channel at a rate $\sim (T_0/2)^2 g^4$ — a vacuum attenuation of light growing as INTENSITY SQUARED, with no accompanying scattered light (the channel is dark) and no frequency conversion signature. DERIVED: the I^2 shape and the darkness. OPEN: the absolute threshold, set by the unpinned vacuum tension density Σ (the same Σ ledger as Prediction 19; QGATE-007).

Test and falsifier. High-intensity laser propagation in vacuum: any observed vacuum attenuation NOT scaling as I^2, or attenuation accompanied by scattered/converted light at the same threshold, falsifies the vertex structure. On the consistency side, existing non-observation of attenuation at petawatt intensities pushes Σ upward, joining Prediction 19's one-parameter ledger — two independent experiments now price the same Σ. Scale-open per the Part IV convention: shape and darkness committed, location open.

A Registered Limit — The Funneling Step (not a prediction)

This paper's discipline requires stating where the mechanism ends as loudly as where it succeeds. The three-session exclusivity campaign (FND-STRAND-022..024, v3.8.0) closed “one quantum, one click” at the classical level -- the measured detector response is a sharp threshold cliff, so a split quantum cannot double-click because half a quantum cannot click at all (half-packet: 0/48; one-quantum split: 0/64 in either region) -- and in doing so SURVEYED the classical engine's jurisdiction: real single photons behind a beamsplitter click exactly one arm at ~50% each, and the engine's split packet clicks neither. The reassembly of a split quantum's full energy at ONE absorption site -- the FUNNELING STEP -- is therefore not performed by any classical local dynamics in this framework's registered detector, a boundary established by measurement rather than argument (benchmarks/foundations/strand_split_packet.py). Consequently: (i) Predictions 11, 23, and 24 are commitments about UNSPLIT quanta and about detector statistics, where the classical engine has jurisdiction; (ii) beamsplitter single-arm statistics, which-path experiments, delayed choice, and the quantum eraser lie beyond the surveyed line and are NOT predicted by the present corpus; (iii) the closure has since been ADOPTED as the framework's third grant (FND-STRAND-025, “the quantum arrives whole”): unit energy transfer at absorption, which with the registered Born-from-channel-energy law DERIVES the single-arm statistics, exact anticoincidence, Fock Binomial counting, and the recovered coherent limit -- with falsifiers armed (partial-quantum deposition; heralded coincidences beyond accidentals; non-Binomial Fock counting) and a standing retirement clause should any local dynamics reproduce these consequences from below. The funneling MECHANISM remains the named open problem, and this limit section remains in force for the dynamics: which-path, delayed choice, and the eraser still lie beyond the surveyed line pending the roadmap's later phases. A framework that documents its edge is falsifiable AT the edge: any demonstration that the classical engine's local dynamics can reproduce single-arm beamsplitter statistics would retire this limit and should be registered against FND-STRAND-024.

NEW PREDICTIONS FROM THE ELECTROMAGNETIC ARC (added 2026-08-09; EM-RECON-029)

Prediction 28 --- The Sign Lock. The collective carrier requires beta in (0, 1/12], so quadratic photon Lorentz violation, at whatever level it exists, must be SUBLUMINAL: high-energy photons arrive late, never early. One bit, zero parameters. KILL: any confirmed superluminal quadratic LIV signal. Maxwell predicts neither sign; quantum-gravity models allow both; the rope allows exactly one.

Prediction 29 --- The Linked Cutoff. One lattice spacing supplies both the dispersion scale E_QG2 and the transparency ceiling E_max, locked in ratio E_QG2/E_max >= sqrt(3) exactly. If quadratic subluminal dispersion is ever measured at scale X, the vacuum must become opaque to photons above X/sqrt(3). No other framework links these observables; a joint measurement is a fingerprint. KILL: a measured dispersion scale together with freely propagating photons above X/sqrt(3). Currently consistent (LHAASO PeV transparency vs the dispersive bound, five orders of headroom).

Prediction 30 --- The Massive Partner. The exact matrix that makes the photon massless predicts its gapped sibling: the relative (optical) branch at E_gap >= ~2e11 GeV. Every accelerator search to date is predicted null; a transverse massive photon-partner below ~1e11 GeV kills the two-branch structure. Maxwell has no partner at any mass.

Prediction 31 --- The One-Number Lock and the Density Obligation. kappa_0 = c/sqrt(eps0 SIGMA): one PVLAS-class measurement of SIGMA fixes the Magnus normalization, every electromagnetic magnitude, and the vacuum medium density simultaneously, with zero remaining freedom. The registered floor SIGMA >= 4e24 J/m^3 implies rho >= 4.5e7 kg/m^3 -- and therefore a QUANTIFIED INTERNAL FALSIFIER: the uniform medium must not gravitate (the induced-gravity route is the registered position); any derivation showing the uniform background sourcing curvature kills the framework on cosmology by ~33 orders. The embarrassment is registered with a number on it, which is this corpus's policy for embarrassments.

Candidate (gated, not yet a prediction) --- Linear optical response of neutral defects: the q-even contact channel implies a linear-in-amplitude driven oscillation of neutral topological defects at optical frequency, where standard physics allows only quadratic polarizability. The composite-neutral coupling requires its own chartered computation before this is asserted; the named confrontation is neutron interferometry in intense optical fields.